JIHWAN KIM

STRATEGY & ANALYTICS | DATA-DRIVEN OPERATIONS | SUSTAINABILITY

Tokyo, Japan | tho3ov@gmail.com | Korean (native), English & Japanese (professional working proficiency)

PROFESSIONAL SUMMARY

Ph.D. oceanographer and analytics professional translating scientific, geospatial, and operational data into commercial and regulatory decisions. Experienced in predictive modeling, digital-product development, compliance analytics, and international stakeholder management across France, Japan, and Asia-Pacific markets. Seeking strategy and transformation roles focused on data-driven growth and operational improvement.

Core capabilities: Strategy & structured problem solving | Predictive modeling & scenario analysis | Data and digital products | Operations & compliance analytics | International stakeholder management

PROFESSIONAL EXPERIENCE

Ocean Engineer, Sustainable Fisheries Management Team — 2025–Present

Collecte Localisation Satellites (CLS) | France & Japan

- Deliver satellite-based fisheries intelligence and fishing-area recommendations to **10✦ commercial clients** and **approximately 150 tuna vessels** across Korea, Japan, Taiwan, the Philippines, and the United States, covering operations in the Pacific, Indian, and Atlantic Oceans.
- Built the **RFMO LL Favorability Map pipeline**, combining RFMO catch and effort records with ocean data to rank fishing areas daily. Developed five models across the Indian and Pacific Oceans and benchmarked the rankings against monthly historical catch rates.
- Developed **ENSO/IOD Catch Alert**, using statistical and machine-learning models to anticipate shifts in tuna purse-seine catches up to six months ahead. Produced seasonal fishing-ground outlooks for fleet planning.
- Contribute to the design and implementation of CATSAT, a satellite-based fisheries intelligence platform, by translating operational needs and predictive models into data requirements, algorithms, and decision-support features.
- Partner with client-facing, scientific, and technical teams in France and Japan to convert vessel-operating needs into analytical workflows, product requirements, and actionable outputs.

Data & Fisheries Science Advisor (Consultant) — 2024–2026

North Pacific Fisheries Commission (NPFC) | Tokyo, Japan

- Advise an intergovernmental fisheries commission on resource management, compliance monitoring, and data governance, presenting evidence-based findings and recommendations to representatives from **7✦ member delegations**.
- Processed and quality-controlled **approximately 9 million VMS records** across three years (approximately 3 million annually) and designed matching algorithms linking VMS, AIS, transshipment declarations, boarding-inspection records, and other operational datasets, identifying **100✦ confirmed data inconsistencies and anomalous cases per year**.
- Developed GIS-based tools to visualize catch-and-effort patterns and support cross-country analysis of North Pacific fisheries, including Pacific saury, chub mackerel, and neon flying squid.
- Produced and presented Fisheries, Transshipment, and VMS overviews at NPFC technical meetings and compiled commission data for the FAO Fisheries and Resources Monitoring System.
- Applied stochastic simulation, statistical forecasting, and deep-learning super-resolution methods to investigate climate-driven changes in fish distribution and strengthen evidence-based resource-management discussions.

Postdoctoral Fellow — 2023–2024

North Pacific Fisheries Commission & Tokyo University of Marine Science and Technology | Tokyo, Japan

- Led quantitative studies connecting basin-scale climate variability and Kuroshio dynamics with changes in Pacific saury and chub mackerel distribution, abundance, and catch performance.
- Developed monitoring concepts and algorithms for IUU-fishing detection by integrating vessel-tracking and air-surveillance information, and proposed measures to improve catch-reporting accuracy.
- Designed and implemented geospatial data products that enabled commission members to compare fisheries activity consistently across jurisdictions and reporting formats.

EARLIER EXPERIENCE & SELECTED PROJECTS

Earth Science Teacher **2018–2019**

Sehwa High School | Seoul, Republic of Korea

Researcher – Korea/Tsushima Strait Observation System **2020–2023**

Seoul National University | Republic of Korea & Japan

- Designed a replacement observation solution, new data-analysis method, and cloud data server to restore submarine-cable voltage monitoring after equipment failure, enabling continued estimation of volume transport through the Korea/Tsushima Strait.

EDUCATION

Ph.D. in Physical Oceanography **2019–2023**

Seoul National University, School of Earth and Environmental Sciences | Seoul, Republic of Korea

- Focus: climate–fisheries forecasting, upper-ocean circulation, machine learning, and geospatial analysis.

B.S. in Earth Science Education **2013–2018**

Seoul National University, College of Education | Seoul, Republic of Korea

REPRESENTATIVE MODELING & ADVISORY ENGAGEMENTS

- Developed stochastic spatio-temporal forecasting and a **multi-head attention U-Net** to improve seasonal fishing-ground outlooks and generate physically consistent super-resolution distributions from coarse data.
- Linked ENSO/IOD, Kuroshio, and Pacific Decadal Oscillation variability to tuna and mackerel catch patterns several months in advance, translating climate signals into operational and management insights.

TECHNICAL & BUSINESS SKILLS

Analytics & technology: Statistical forecasting, regression, scenario analysis, spatio-temporal modeling, machine learning, deep learning, geospatial analysis; Python, MATLAB, SQL/MySQL, GIS, dashboards, and data visualization

Additional tools: Working knowledge of Julia and Fortran; web-application prototyping, database design, and requirements definition

Consulting-relevant strengths: Problem structuring, evidence synthesis, executive-ready reporting, requirements definition, cross-functional collaboration, international stakeholder communication

SELECTED PUBLICATIONS & RECOGNITION

- Published peer-reviewed research in **Scientific Reports and Frontiers in Marine Science** and presented fisheries forecasting, vessel-monitoring, and climate–ecosystem findings to NPFC committees, PICES, AGU, and the Korea Oceanographic Society.
- **2nd Prize**, JOISS Marine Science Big Data Competition (2021 and 2022; UN Ocean Decade Activity); Succeeding Generations of Fundamental Sciences Fellowship, Seoul National University (full scholarship, 2019–2020).